Algebraic Number Theory

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Disclaimer

This will be a rewrite of the course of Pr. P. STEVENHAGEN, it will be far from the explanations or the accompanied textbook of the course.

To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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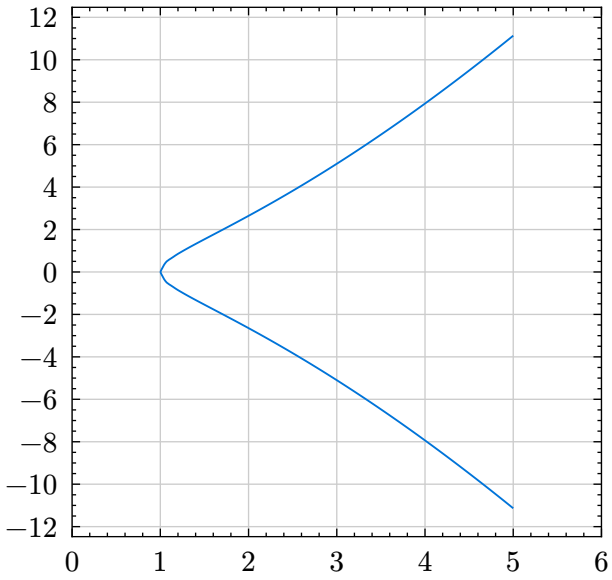
Chapter 1

Introduction

In this part, we motivate the usage of algebraic number theory using elliptic curves. Consider the following elliptic curve

$$E : y^2 = x^3 - 1$$



We would like to prove the following theorem

Theorem: *The only integral solution to E is $(0, 1)$.*

Proof. By rewriting our equation into $x^3 = y^2 + 1$, we can think of writing the RHS as a product, that is, by considering on the ring $\mathbb{Z}[i] \subset \mathbb{Q}[i]$, which is a unique factorization domain, we have that $y^2 + 1 = (y + i)(y - i)$.

Now we prove that $y + i$ and $y - i$ are coprime. Suppose that there is a prime π such that $\pi \mid y + i$ and $\pi \mid y - i$, then $\pi \mid y + i - y + i = 2i$ and since $i \in (\mathbb{Z}[i])^\star$ then necessarily $\pi \mid 2$. In $\mathbb{Z}[i]$, 2 is factorized as $(1 + i)(1 - i) = -i(1 + i)^2$, thus π is necessarily $i + 1$ times some unit. By reducing mod 4 both x^3 and $y^2 + 1$, we obtain that $x^3 \equiv 0, 1 \bmod 4$ and $y^2 + 1 \equiv 1, 2 \bmod 4$ thus $x^3 \equiv 1 \bmod 4$ so x is odd and $y^2 \equiv 0 \bmod 4$ so y is even. Given that $i + 1 \mid \pi$ then necessarily we have $1 + i \mid y + i$ which is impossible since y is even. Thus, $y + i$ and $y - i$ are coprime. $\square$

Chapter 2

p-adic Numbers

Consider $K/\mathbb{Q}$ a number field, to be able to understand the arithmetic of K, we want to understand for every p prime the decomposition

$$p \cdot O_k = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}$$

for now the way we can do it is using the Kummer-Dedekind criterion stating that if $\alpha \in O_k$ such that $K = \mathbb{Q}(\alpha)$ and $p \nmid [O_k : \mathbb{Z}[\alpha]]$, then we can understand $p \cdot O_k$ from the factorization modulo p of $\min_\alpha(X) \in \mathbb{Z}[x]$.

In particular, if $\overline{\min_\alpha}(X) = \prod_{i=1}^t \overline{g_i}(X)^{e_i}, \overline{g_i}(X) \in \mathbb{F}_{p[X]}$ irreducible, then we obtain the following factorization $p \cdot O_k = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}$ where $\mathfrak{p}_i = (p, g_i(\alpha))$ where $g_i \in \mathbb{Z}[X]$ is any lift of $\overline{g_i}(x)$.

The problem happens when $p \mid [O_k : \mathbb{Z}[\alpha]]$ for any $\alpha \in O_k$. The aim of this part of the course is to get around this issue.

We will see that we can build for any prime p a field $\mathbb{Q}_p/\mathbb{Q}$ such that $K \otimes_\mathbb{Q} \mathbb{Q}_p = \prod_{i=1}^t K_{\mathfrak{p}_i}$ with $K_{\mathfrak{p}_i}/\mathbb{Q}_p$ finite extension $\left[K_{\mathfrak{p}_i} : \mathbb{Q}_p\right] = e_i f_i$ and $p \cdot O_k = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}$.

Note: *If $K = \mathbb{Q}(\alpha) = \mathbb{Q}[X]/(\min_\alpha(X))$ then*

$$K \otimes_\mathbb{Q} \mathbb{Q}_p = \mathbb{Q}[X]/(f(X)) \otimes_\mathbb{Q} \mathbb{Q}_p$$
$$\cong \mathbb{Q}_p[X]/(f(X)) \cong \prod_{i=1}^t \mathbb{Q}_p[X]/(h_i(X))$$

by the CRT, $f(x) = h_1(X) \cdots h_t(X)$ in $\mathbb{Q}_p[X]$, $h_{i(X)}$ irreducible over $\mathbb{Q}_p[X]$. If instead of $\mathbb{Q}_p$ we take $\mathbb{R}$, we have

$$K \otimes_\mathbb{Q} \mathbb{R} \cong \mathbb{R}[X]/(f(X)) \cong \prod_{i=1}^r \frac{\mathbb{R}[X]}{x - \alpha_i} \times \prod_{j=1}^s \frac{\mathbb{R}[X]}{h_j(X)} \cong \mathbb{R}^r \times \mathbb{C}^s$$

where $f(X) = (X - \alpha_1) \cdots (X - \alpha_r) h_1(X) \cdots h_s(X)$, $h_i \in \mathbb{R}[X]$ irreducible of degree 2, r is the number of real embeddings and s the number of pairs of complex conjugate embeddings.

Now we go on how to build $\mathbb{R}$ starting from $\mathbb{Q}$. On $\mathbb{Q}$ we have the Euclidean absolute value

$$\Phi_\infty : \mathbb{Q} \to \mathbb{Q}_{\geq 0}$$
$$x \mapsto |x|$$

and $\mathbb{R}$ is the completion of $\mathbb{Q}$ with respect to Φ_∞. We define R as the set of Cauchy sequences with respect to Φ_∞ and m as the set of sequences that tends to zero.

Exercise: *Prove that R with component-wise addition and multiplication form a ring, and m is an ideal of R thus R/m is a ring and more specifically it is a field, which we denote as $\mathbb{R} = R/m$.*

If you look at the exercise you will see that the only thing you need in order to have that R/m is a field is a function $\Phi : \mathbb{Q} \to \mathbb{Q}_{\geq 0}$ that satisfies:

- $\Phi(x) = 0 \Leftrightarrow x = 0$.
- $\Phi(xy) = \Phi(x)\Phi(y)$.
- $\Phi(x + y) \leq C \cdot \max(\Phi(x), \Phi(y))$

2.1. p-adic Numbers

Fix a prime $p \in \mathbb{Z}$, we defined the order function as follows

$$\mathrm{ord}_p : \mathbb{Q} \setminus \{0\} \to \mathbb{Z} \qquad \frac{a}{b} \mapsto \mathrm{ord}_p(a) - \mathrm{ord}_p(b)$$

with $a, b \in \mathbb{Z}$, $\gcd(a, b) = 1$, where if $a \in \mathbb{Z}$ $\mathrm{ord}_p(a) = n$, $p^n \mid a$ and $p^{n+1} \nmid a$.

Example: For $p = 5$:

- $\mathrm{ord}_5(7) = 0$.
- $\mathrm{ord}_5(30) = \mathrm{ord}_5(5 \cdot 6) = 1$.
- $\mathrm{ord}_5\left(-\frac{4}{125}\right) = \mathrm{ord}_5(-4) - \mathrm{ord}_5(5^3) = -3$.

Note: *We extend* ord_p *to* $\mathbb{Q}$ *by defining* $\mathrm{ord}_p(0) = \infty$.

Definition (p-adic Norm): *Let* $c \in (0,1)$, *then we can define*

$$\Phi_p : \mathbb{Q} \to \mathbb{R}_{\geq 0} \qquad x \mapsto c^{\mathrm{ord}_p(x)}$$

Exercise:
- $\Phi_p(x) = 0 \Leftrightarrow x = 0$.
- $\Phi_p(xy) = \Phi_p(x)\Phi_p(y)$.
- $\Phi_p(x+y) \leq \max\big(\Phi_p(x), \Phi_p(y)\big)$.

Definition: *We take* $\mathbb{Q}_p = R/m$ *where* R *is the set of Cauchy sequences of* $\mathbb{Q}$ *with respect to* Φ_p, *and* m *is the set of sequences of* $\mathbb{Q}$ *that tend to* 0 *with respect to* Φ_p.

In the real numbers, any $x \in \mathbb{R}$ is written as $x = \pm \sum_{n=k}^{\infty} a_n \cdot 10^{-n}$ for some $k \in \mathbb{Z}$ and $a_n \in \{0, \cdots, 9\}$. Similarly, over $\mathbb{Q}_p$ every element can be written as $\sum_{n=k}^{\infty} a_n p^n$ with $k \in \mathbb{Z}$ and $a_n \in \{0, \ldots, p-1\}$.

Lemma: *Define* $\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid \Phi_p(x) \leq 1\}$
- $\mathbb{Z}_p$ *is a ring.*
- $\mathbb{Z} \subset \mathbb{Z}_p$.
- $\mathbb{Z}_p$ *has just one non-zero prime ideal given by* $p\mathbb{Z}_p \subset \mathbb{Z}_p$, *we have that* $\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z}$.

Proposition: *For any* $x \in \mathbb{Q}_p$, *there exists a unique sequence* $(a_n)_{n\in\mathbb{N}} \subset \{0, \ldots, p-1\}^{\mathbb{N}}$ *such that* $x = \sum_{n=0}^{\infty} a_n p^n$.

Proof.

Let $x \in \mathbb{Q}_p$.
- There exists $k \in \mathbb{Z}_{\geq 0}, p^k x \in \mathbb{Z}_p$. (Exercise: $\Phi_p(p^k x) = c^k \Phi_p(x) \leq 1$.)
- Show that for $x \in \mathbb{Z}_p, x = \sum_{n=0}^{\infty} a_n p^n$.
- Take $a_0 \in [\![0, p-1]\!], x \equiv a_0 \bmod p$ thus $x - a_0 = px_1$ for some $x_1 \in \mathbb{Z}_p$. Take a_1 such that $x_1 \equiv a_1 \bmod p$ thus $x - a_0 - pa_1 - \ldots - p^n a_n \equiv 0 \bmod p^n$ hence $x = \sum_{n=0}^{\infty} a_n p^n$. □

2.2. Factorizing Polynomials Over $\mathbb{Q}_p[X]$

If $\alpha \in \mathbb{Q}_p$ is a root of $f(X)$ then $f(X) = (X - \alpha)g(X)$. Start with $f(X) \in \mathbb{Z}[X]$.
- If $f(X) \equiv 0 \bmod p$ has no solution, then we don't have any solution over $\mathbb{Z}_p$.
- If $f(X) \equiv 0 \bmod p$ has a solution, we take $\alpha_1 \in \mathbb{Z}_p, f(\alpha_1) \equiv 0 \bmod p$.
- If $f(X) \equiv 0 \bmod p^2$ has a solution, we take $\alpha_2 \in \mathbb{Z}_p$ such that $f(\alpha_2) \equiv 0 \bmod p^2$ and $\alpha_2 \equiv \alpha_1 \bmod p$.
- By continuing this process, assume that $\exists \alpha_n \in \mathbb{Z}_p$ such that $f(\alpha_n) \equiv 0 \bmod p^n$ and $\alpha_n \equiv \alpha_{n-1} \bmod p^{n-1}$.

Therefore, $(a_n)_{n\in\mathbb{N}}$ is a sequence of elements in $\mathbb{Z}_p$. By $\alpha_n \equiv \alpha_{n-1} \bmod p^{n-1}$, we get that the sequence $(\alpha_n)_{n\in\mathbb{N}}$ is Cauchy, thus there exists $\alpha = \lim_{n\to\infty} \alpha_n \in \mathbb{Z}_p$, and from the condition $f(\alpha_n) \equiv 0 \bmod p^n$ we get $\lim_{n\to\infty} f(\alpha_n) = 0 = f(\alpha)$.

Lemma (Hensel): *Let* $f(X) \in \mathbb{Z}_p[X]$ *be a polynomial, let* $\alpha_0 \in \mathbb{Z}_p$ *with*

$$\Phi_p(f(\alpha_0)) < \Phi_p(f'(\alpha_0))^2$$

or equivalently

$$\mathrm{ord}_p(f(\alpha_0)) > 2 \cdot \mathrm{ord}_P(f'(\alpha_0))$$
$$\Rightarrow \mathrm{ord}_p(f(\alpha_0)) > 0 \Rightarrow f(\alpha_0) \equiv 0 \bmod p$$

Then $\exists \alpha \in \mathbb{Z}_p$ *such that* $f(\alpha) = 0$ *and* $\Phi_p(\alpha - \alpha_0) \leq \Phi_p\left(\frac{f(\alpha_0)}{f'(\alpha_0)}\right)$.

Note: *If* $f(X) \in \mathbb{Z}[X]$ *and* $\alpha_0 \in \mathbb{Z}$ *such that* $f(\alpha_0) \equiv 0 \bmod p$ *and* $f'(\alpha_0) \not\equiv 0 \bmod p$. *Thus, there exists* $\alpha \in \mathbb{Z}_p$ *such that* $f(\alpha) = 0$.

In Fact: $f(\alpha_0) \equiv 0 \bmod p \Rightarrow \mathrm{ord}_p(f(\alpha_0)) \geq 1 \Rightarrow \Phi_p(f(\alpha_0)) \leq c$. $f'(\alpha_0) \not\equiv \bmod p \Rightarrow \mathrm{ord}_p(f'(\alpha_0)) = 0 \Rightarrow \Phi_p(f'(\alpha_0)) = 1 \Rightarrow \Phi_p(f(\alpha_0)) < \Phi_p(f'(\alpha_0))^2 = 1 \Rightarrow \exists \alpha \in \mathbb{Z}_p, f(\alpha) = 0$ *by Hensel's lemma.*

Example:

- $K = \mathbb{Q}\left(\sqrt{-13}\right) = \mathbb{Q}[X]/(X^2+13)$, $f(X) = X^2+13$, $f'(X) = 2X$ and $p = 7$. Then $f(1) = 14 \equiv 0 \bmod 7$ and $f'(1) = 2 \not\equiv 0 \bmod 7$ then $\exists \alpha \in \mathbb{Z}_7, f(\alpha) = 0 \Rightarrow X^2+13 = (X-\alpha)(X+\alpha)$ and we have

$$K \otimes_{\mathbb{Q}} \mathbb{Q}_7 \equiv \frac{\mathbb{Q}_7[X]}{(X^2+13)} \equiv \frac{\mathbb{Q}_7[X]}{(X-\alpha)} \times \frac{\mathbb{Q}_7[X]}{(X+\alpha)} \equiv \mathbb{Q}_7 \times \mathbb{Q}_7$$

> **Exercise 2.2.1:** *Compare it with $7 \cdot O_k$ using Kummer-Dedekind criteria.*

- When is $\alpha \in \mathbb{Z}_2$ a square ?
  - $\alpha = 13, f(X) = X^2 - 13, f'(X) = 2X$
    - $\bmod 2 : f(1) \equiv -12 \equiv 0 \bmod 2, f'(1) \equiv 2 \equiv 0 \bmod 2$.
    - $\bmod 8 : f(X) = X^2 - 13$ has no solution thus there is no solution over $\mathbb{Z}_2$.
  - $\alpha = 17, f(X) = X^2 - 17, f'(X) = 2X$
    - $\bmod 2 : f(1) \equiv -16 \equiv 0 \bmod 2, f'(1) \equiv 2 \equiv 0 \bmod 2$.
    - $\bmod 8 : f(1) \equiv 0 \bmod 8 \Rightarrow \Phi_p(f(1)) \leq c^3, f'(1) \equiv 2 \not\equiv 0 \bmod 8$ then $\Phi_p(f'(1)) = c, \exists \alpha \in \mathbb{Z}_2, f(\alpha) = 0$.

- Consider $f(X) = X^3 - X^2 - 2X - 8$ then $f'(X) = 3X^2 - 2X - 2$, we want to know how to factor $f(X)$ in $\mathbb{Q}_2[X]$
  1. $f(X) \equiv 0 \bmod 16$ has 5 solutions which are $4, 6, 7, 12, 14$.
     1. for $\alpha = 7, \Phi_2(f(7)) = c^3, \Phi_2(f'(7)) = 1$ then there exists $\alpha \in \mathbb{Z}_2$ such that $f(\alpha_1) = 0$.
     2. for $\alpha = 6, 12$ we have $\varphi_2(f(\alpha)) = c^5, \varphi_2(f'(\alpha)) = c$ thus $\exists \alpha_2, \alpha_3 \in \mathbb{Z}_2, f(\alpha_2) = f(\alpha_3) = 0$.
     3. Over $\mathbb{Q}_2$ we have 3 different roots so

$$f(X) = (X-\alpha_1)(X-\alpha_2)(X-\alpha_3)$$

     and hence $K \otimes \mathbb{Q}_2 \equiv Q_2 \times Q_2 \times Q_2 \Rightarrow 2 \cdot O_k = \mathfrak{p}_1\mathfrak{p}_2\mathfrak{p}_3$.

  Assume that $\exists O \in O_k$ st that $2 \nmid [O_k : \mathbb{Z}[O]]$, we can read the factorization of $\mathbb{Z}$ from the real modulo $\mathbb{Z}$ of $\min_O(X)$. However, $\min_O(x)$ cannot have three different linear factors over $\mathbb{F}_2[X]$ (X and $X+1$ are the only linear factors).

> **Lemma:** *Suppose that $f(X) \in \mathbb{Z}_p[X]$ is monic, if modulo $\overline{f}(T) = \overline{g}(T) \cdot \overline{h}(T) \bmod p$ with $\overline{g}(T), \overline{h}(T) \in \mathbb{F}_p[T]$ coprime, then $\exists g(T), h(T) \in \mathbb{Z}_p[T]$ such that $f(T) = g(T) \cdot h(T)$ with $\deg(g) = \deg(\overline{g}), g(T) \equiv \overline{g}(T) \bmod p, \deg(h) = \deg\left(\overline{h}\right), h(T) \equiv \overline{h}(T) \bmod p$.*

Example:

- The coprime assumption cannot be removed, by taking $f(T) = T^2 - 3$ over $\mathbb{Z}_3[T]$. We have $\overline{f}(T) = T^2 = T \cdot T$ but $f(T)$ is irreducible in $\mathbb{Z}_3[T]$ (show that it has no solution modulo 9).
- $f(X) = X^6 - 6X^5 - 8X^4 - 6X - 10$. By reducing on $\mathbb{Z}_3$ we have $\overline{f}(X) = X^6 + X^4 + X^2 - 1 = (X^3 - X + 1)(X^3 - X - 1) = \overline{g}(X)\overline{h}(X)$, then $\overline{g}(X), \overline{h}(X)$ are coprime, and they are irreducible. Thus, by Hensel's lemma, we have that $f(X) = g(X)h(X)$ with $\deg(g) = \deg(H) = 3$ with $g(X)$ and $h(X)$ are irreducible since they are of degree 3 and they don't have any root.